

If you use an online resource to solve any problem, please appropriately cite that source; this includes any AI agent/tool such as ChatGPT, Claude, Gemini, etc.

- All hw should be uploaded to canvas as a *pdf*. Make sure that if you scan your handwritten notes that they are legible and appropriately oriented.
- Homework is always due on Wednesday after its released at 11:59PM; however, submitting by Monday at 11:59PM will get you extra credit.
- All homework will be self-graded, with the exception of one problem (selected uniformly at random) that will be graded.
- Self-grading is required to get credit and it is a way to get full credit for your homework (up to any mistakes on the random grading of one problem).
- Bonus problems are not required for credit, but they are worth extra credit.

Definitions & Facts. Recall that for a linear system of the form $Ax = b$,

- There exists a solution if $b \in \text{Range}(A)$.
- The solution is unique if A has full column rank, and in particular, the least squares solution is given by $x = (A^\top A)^{-1} A^\top b$.
- If A does not have full column rank, then there are infinitely many solutions.

Problems

Problem 1. (Casting a problem as a least squares problem.) Consider the data points $(x_i, y_i) \in \mathbb{R} \times \mathbb{R}$:

$$\{(x_i, y_i)\}_{i=1}^4 = \{(1, 1), (2, 0), (-1, 2), (0, -1)\}.$$

We aim to determine a real polynomial of degree two that best fits this data, where the polynomial is of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$, and $\theta = (\theta_0, \theta_1, \theta_2)^\top \in \mathbb{R}^3$. Note that there are more data points than there are unknown coefficients, so this problem is overdetermined.

- Write the objective function as a sum of norms defined in terms of the coefficients.
- Write out the matrix-vector equation that defines the problem (i.e., what is X and y ?)
- Solve the problem using the normal equations.

Solution.

$$(\theta_0 + \theta_1 + \theta_2 - 1)^2 + (\theta_0 + 2\theta_1 + 4\theta_2 - 0)^2 + (\theta_0 - \theta_1 + \theta_2 - 2)^2 + (\theta_0 + \theta_2 - (-1))^2.$$

The matrix-vector equation that defines the problem is:

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & -1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad y = \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \end{bmatrix}.$$

Now we can solve the problem using the normal equations:

$$\theta = (X^\top X)^{-1} X^\top y.$$

$$\theta = \left(\underbrace{\begin{bmatrix} 4 & 2 & 6 \\ 2 & 6 & 8 \\ 6 & 8 & 18 \end{bmatrix}}_{X^\top X} \right)^{-1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & 0 \\ 1 & 4 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 6 \\ 2 & 6 & 8 \\ 6 & 8 & 18 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{5} \\ -\frac{9}{10} \\ \frac{1}{2} \end{bmatrix}.$$

Therefore, the polynomial that best fits the data is:

$$y = \frac{1}{5} + \frac{9}{10}x - \frac{1}{2}x^2.$$

Problem 2. (Definition and basic computation.)

Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}.$$

- (a) Write down the definition of the characteristic polynomial of a square matrix A .
- (b) Compute the characteristic polynomial $p_A(\lambda)$ of A .
- (c) Find the eigenvalues of A .

Solution.

(a) By definition,

$$p_A(\lambda) = \det(\lambda I - A).$$

(b)

$$p_A(\lambda) = \det \begin{bmatrix} \lambda - 2 & -1 \\ 0 & \lambda - 3 \end{bmatrix} = (\lambda - 2)(\lambda - 3).$$

(c) The eigenvalues are $\lambda_1 = 2$, $\lambda_2 = 3$.

Problem 3. (Characteristic polynomial and trace/determinant.)

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

- (a) Compute the characteristic polynomial $p_A(\lambda)$ symbolically.
- (b) Express $p_A(\lambda)$ in terms of $\text{tr}(A)$ and $\det(A)$.
- (c) Verify that the sum of eigenvalues equals $\text{tr}(A)$ and the product equals $\det(A)$.

Solution.

(a)

$$p_A(\lambda) = \det(\lambda I - A) = \det \begin{bmatrix} \lambda - a & -b \\ -c & \lambda - d \end{bmatrix} = \lambda^2 - (a + d)\lambda + (ad - bc).$$

The determinant of a 2×2 matrix $\begin{bmatrix} p & q \\ r & s \end{bmatrix}$ is $ps - qr$, so

$$p_A(\lambda) = (\lambda - a)(\lambda - d) - (-b)(-c) = (\lambda - a)(\lambda - d) - bc.$$

Expanding,

$$p_A(\lambda) = \lambda^2 - (a + d)\lambda + (ad - bc).$$

(b) The *trace* of A is $\text{tr}(A) = a + d$, and the *determinant* of A is $\det(A) = ad - bc$. Substituting these into the formula above gives the compact expression

$$p_A(\lambda) = \lambda^2 - (\text{tr } A)\lambda + \det(A).$$

This shows that for any 2×2 matrix, the coefficient of λ is the negative of the trace, and the constant term is the determinant.

(c) If λ_1 and λ_2 are the eigenvalues of A , then by definition they are the roots of $p_A(\lambda) = 0$. For a quadratic

$$p_A(\lambda) = \lambda^2 - (\text{tr } A)\lambda + \det(A),$$

we know from the factorization of quadratics that

$$p_A(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2.$$

Matching coefficients gives

$$\lambda_1 + \lambda_2 = \text{tr}(A), \quad \lambda_1\lambda_2 = \det(A).$$

Hence, the trace of A equals the sum of its eigenvalues, and the determinant equals their product.

Problem 4. ((VMLS 12.15) Minimizing a squared norm plus an affine function.)

Consider

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|^2 + c^\top x + d,$$

where $A \in \mathbb{R}^{m \times n}$ has linearly independent columns, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, $d \in \mathbb{R}$.

Show that the objective can be written

$$\|Ax - b\|^2 + c^\top x + d = \|Ax - b + f\|^2 + g$$

for some $f \in \mathbb{R}^m$ and constant $g \in \mathbb{R}$, and identify a valid choice of f and g . Hence the problem reduces to an ordinary least-squares problem.

Hint: Express the normal squared term on the right-hand side as $\|(Ax - b) + f\|^2$ and expand. Then argue the equality above holds provided that $2A^\top f = c$. One possible choice is $f = \frac{1}{2}(A^\dagger)^\top c$. (You must verify these steps.)

Solution. Expand the right-hand side:

$$\|Ax - b + f\|^2 = \|Ax - b\|^2 + 2f^\top(Ax - b) + \|f\|^2.$$

Equality with the left-hand side for all x holds iff the terms linear in x match:

$$c^\top x \equiv 2f^\top Ax \iff c = 2A^\top f.$$

With any f satisfying $2A^\top f = c$, set

$$g = \|f\|^2 - 2f^\top b + d,$$

and then $\|Ax - b\|^2 + c^\top x + d = \|Ax - b + f\|^2 + g$.

Because A has full column rank, $A^\top : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is surjective, so such f exists. A convenient choice is

$$f = \frac{1}{2} (A^\dagger)^\top c = \frac{1}{2} A (A^\top A)^{-1} c,$$

which satisfies $A^\top f = \frac{1}{2}c$.

Therefore the original problem is equivalent to minimizing $\|Ax - (b - f)\|$, an ordinary least-squares problem whose solution is

$$\hat{x} = A^\dagger(b - f).$$

(Any constant g is irrelevant to the minimizer.)

Problem 5. ((VMLS 12.14) Recursive least squares.) In some applications of least squares the rows of the coefficient matrix A become available sequentially. We wish to solve the growing least-squares problems with data

$$A^{(k)} = \begin{bmatrix} a_1^\top \\ \vdots \\ a_k^\top \end{bmatrix} \in \mathbb{R}^{k \times n}, \quad b^{(k)} = \begin{bmatrix} b_1 \\ \vdots \\ b_k \end{bmatrix} \in \mathbb{R}^k, \quad k = n, n+1, \dots, m,$$

assuming the columns of $A^{(n)}$ are linearly independent (hence those of $A^{(k)}$ are independent for all $k \geq n$) and $m \gg n$. Define

$$G^{(k)} = (A^{(k)})^\top A^{(k)} \in \mathbb{R}^{n \times n}, \quad h^{(k)} = (A^{(k)})^\top b^{(k)} \in \mathbb{R}^n.$$

(a) Show that the least-squares estimate $\hat{x}^{(k)}$ satisfies

$$\hat{x}^{(k)} = (G^{(k)})^{-1} h^{(k)}, \quad k = n, \dots, m.$$

(b) Show the recursions

$$G^{(k+1)} = G^{(k)} + a_{k+1} a_{k+1}^\top, \quad h^{(k+1)} = h^{(k)} + b_{k+1} a_{k+1}, \quad k = n, \dots, m-1.$$

Solution.

(a) The least-squares normal equations for the k th problem are $(A^{(k)})^\top A^{(k)} \hat{x}^{(k)} = (A^{(k)})^\top b^{(k)}$. Since $A^{(k)}$ has full column rank, $G^{(k)}$ is invertible, giving $\hat{x}^{(k)} = (G^{(k)})^{-1} h^{(k)}$.

(b) By construction,

$$G^{(k+1)} = (A^{(k+1)})^\top A^{(k+1)} = [(A^{(k)})^\top \quad a_{k+1}] \begin{bmatrix} A^{(k)} \\ a_{k+1}^\top \end{bmatrix} = G^{(k)} + a_{k+1} a_{k+1}^\top.$$

Similarly,

$$h^{(k+1)} = (A^{(k+1)})^\top b^{(k+1)} = [(A^{(k)})^\top \quad a_{k+1}] \begin{bmatrix} b^{(k)} \\ b_{k+1} \end{bmatrix} = h^{(k)} + b_{k+1} a_{k+1}.$$

Problem 6. (Projection Matrix onto a Subspace.) Let $X \in \mathbb{R}^{m \times n}$ have linearly independent columns, representing n basis vectors for a subspace

$$\mathcal{C} := \text{col}(X) = \{z \mid \exists \theta \in \mathbb{R}^n, z = X\theta\} \subseteq \mathbb{R}^m.$$

For any vector $y \in \mathbb{R}^m$, we define its projection onto $\mathcal{C}$ as

$$\hat{y} := Py, \quad \text{where } P = X(X^\top X)^{-1}X^\top.$$

(a) Show that P is symmetric and idempotent; that is,

$$P^\top = P, \quad P^2 = P.$$

(b) Show that $\hat{y} = Py$ is the unique vector in $\text{col}(X)$ that minimizes

$$\|y - X\theta\|_2^2.$$

(Hint: use the normal equations $X^\top X\hat{\theta} = X^\top y$.)

(c) Verify that the residual $r := y - \hat{y}$ satisfies $X^\top r = 0$.

(d) Let

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad y = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

Compute P , $\hat{y}$, and the residual $r = y - \hat{y}$ explicitly. Check that r is orthogonal to the columns of X .

Solution.

(a) Symmetry and idempotence. Since $(X^\top X)^{-1}$ is symmetric,

$$P_X^\top = (X(X^\top X)^{-1}X^\top)^\top = X(X^\top X)^{-1}X^\top = P_X.$$

Also,

$$P_X^2 = X(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1}X^\top = X(X^\top X)^{-1}X^\top = P_X.$$

(b) Least-squares minimizer. Minimizing $\|y - X\theta\|_2^2$ gives the normal equations $X^\top X\hat{\theta} = X^\top y$. Then $\hat{y} = X\hat{\theta} = X(X^\top X)^{-1}X^\top y = P_X y$. Uniqueness follows from $\text{rank}(X) = n$.

(c) Orthogonality of residual. $r = y - \hat{y} = y - P_X y$. Then

$$X^T r = X^T (I - P_X) y = X^T y - X^T X (X^T X)^{-1} X^T y = 0.$$

Hence $r \perp \text{col}(X)$.

(d) Numerical example. Compute step by step:

$$X^T X = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad (X^T X)^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Then

$$P_X = X (X^T X)^{-1} X^T = \frac{1}{3} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$$

Therefore

$$\hat{y} = P_X y = \frac{1}{3} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.67 \\ 1.33 \\ 0.33 \end{bmatrix}.$$

Then

$$r = y - \hat{y} = \begin{bmatrix} 0.33 \\ -0.33 \\ -0.33 \end{bmatrix}, \quad X^T r = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.33 \\ -0.33 \\ -0.33 \end{bmatrix} = 0.$$

This verifies $r \perp \text{col}(X)$.

Problem 7. (Least Squares and QR Factorization.) Suppose X is an $m \times n$ matrix with linearly independent columns, and $X = QR$ is its QR factorization, where $Q \in \mathbb{R}^{m \times n}$ has orthonormal columns ($Q^T Q = I_n$), and $R \in \mathbb{R}^{n \times n}$ is upper triangular. For a given vector $b \in \mathbb{R}^m$, the least-squares solution $\hat{\theta}$ minimizes $\|X\theta - b\|_2$. The vector $X\hat{\theta}$ is the projection of b onto the column space of X .

(a) Show that $X\hat{\theta} = QQ^T b$. (The matrix QQ^T is called the projection matrix.)

(b) Show that

$$\|X\hat{\theta} - b\|_2^2 = \|b\|_2^2 - \|Q^T b\|_2^2.$$

(This is the square of the distance between b and the closest linear combination of the columns of X .)

Solution.

(a) The least-squares problem is

$$\min_x \|Ax - b\|_2^2 = \min_x \|QRx - b\|_2^2.$$

Since Q has orthonormal columns, we can decompose b as

$$b = QQ^T b + (I - QQ^T)b,$$

where $QQ^\top b$ lies in $\text{col}(Q) = \text{col}(A)$ and $(I - QQ^\top)b$ is orthogonal to that subspace. Minimizing $\|QRx - b\|_2^2$ is equivalent to minimizing

$$\|Q^\top(QRx - b)\|_2^2 = \|Rx - Q^\top b\|_2^2.$$

The minimizer satisfies $Rx = Q^\top b$, hence

$$\hat{x} = R^{-1}Q^\top b, \quad A\hat{x} = QR\hat{x} = Q(Q^\top b) = QQ^\top b.$$

Thus $A\hat{x}$ is indeed the projection of b onto $\text{col}(A)$.

(b) The residual is

$$r = b - A\hat{x} = (I - QQ^\top)b.$$

Because $QQ^\top$ is an orthogonal projector, the two components are orthogonal:

$$(QQ^\top b)^\top (I - QQ^\top)b = 0.$$

By the Pythagorean theorem,

$$\|b\|_2^2 = \|QQ^\top b\|_2^2 + \|(I - QQ^\top)b\|_2^2.$$

Therefore,

$$\|A\hat{x} - b\|_2^2 = \|r\|_2^2 = \|b\|_2^2 - \|QQ^\top b\|_2^2.$$

Finally, since Q has orthonormal columns, $\|QQ^\top b\|_2 = \|Q^\top b\|_2$, giving

$$\boxed{\|A\hat{x} - b\|_2^2 = \|b\|_2^2 - \|Q^\top b\|_2^2.}$$

Problem 8. (Minimum-Norm Trick for Linear Systems.) We want to find the smallest-norm vector θ that satisfies the linear system

$$X\theta = y,$$

where $X \in \mathbb{R}^{n \times d}$ has full row rank ($n < d$) and the system is consistent. This problem is underdetermined and can be cast as the following optimization problem:

$$\min_{\theta} \|\theta\|_2 \quad \text{s.t. } X\theta = y.$$

(a) Instead of solving the constrained problem directly, consider the *augmented objective*

$$J(\theta) = \frac{1}{2}\|\theta\|_2^2 + \lambda^\top(y - X\theta),$$

where $\lambda \in \mathbb{R}^m$ is a fixed vector.

Compute $\nabla_{\theta} J(\theta)$ and find θ that minimizes $J(\theta)$ in terms of λ .

(b) Substitute your result into the constraint $X\theta = y$ and solve for λ in terms of X and y .

(c) Substitute λ back to get an explicit expression for θ . Show that

$$\hat{\theta} = X^\top (XX^\top)^{-1} y.$$

(d) Explain why this $\hat{\theta}$ is the same as the solution to

$$\min_{\theta} \|\theta\|_2 \quad \text{s.t. } X\theta = y.$$

(Hint: do this in steps. First, first consider any θ that satisfies $X\theta = y$. Then, show that θ can be written as $\theta = \theta_0 + v$, where θ_0 is a particular solution and v is in the null space of X . Finally, by expanding the norm $\|\theta\|_2^2 = \|\theta_0 + v\|_2^2$, show that the minimum norm solution is the one that minimizes $\|\theta\|_2$ over all θ that satisfy $X\theta = y$.)

(e) Let

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Compute $\hat{\theta}$ explicitly and check that $X\hat{\theta} = y$ and that $\hat{\theta}$ has the smallest norm among all feasible solutions.

Solution.

(a) Differentiate with respect to θ :

$$\nabla_{\theta} J(\theta) = \theta - X^T \lambda.$$

Setting $\nabla_{\theta} J = 0$ gives

$$\boxed{\theta = X^T \lambda.}$$

(b) Enforce the constraint $X\theta = y$:

$$X(X^T \lambda) = XX^T \lambda = y \quad \Rightarrow \quad \lambda = (XX^T)^{-1} y.$$

(c) Substitute back:

$$\hat{\theta} = X^T (XX^T)^{-1} y.$$

This vector satisfies $X\hat{\theta} = y$.

(d) Interpretation: $J(\theta)$ penalizes large $\|\theta\|$ while $\lambda^T(y - X\theta)$ enforces the equality constraint. Minimizing J over θ implicitly gives the smallest $\|\theta\|$ that satisfies $X\theta = y$.

Algebraic proof of minimum norm. Let $\hat{\theta} := X^T (XX^T)^{-1} y$. First, $\hat{\theta}$ is feasible:

$$X\hat{\theta} = XX^T (XX^T)^{-1} y = y.$$

Now let $\mathcal{S} := \{\theta \in \mathbb{R}^n : X\theta = y\}$ be the solution set. Fix any $\theta \in \mathcal{S}$. Since $X\hat{\theta} = y$ also, we have

$$X(\theta - \hat{\theta}) = 0 \quad \Rightarrow \quad \theta = \hat{\theta} + v, \quad \text{for some } v \in \text{Null}(X).$$

We claim $\hat{\theta} \perp \text{Null}(X)$. Indeed, for any $v \in \text{Null}(X)$,

$$\hat{\theta}^T v = (X^T (XX^T)^{-1} y)^T v = y^T (XX^T)^{-1} Xv = y^T (XX^T)^{-1} 0 = 0.$$

Therefore, for any feasible $\theta = \hat{\theta} + v$,

$$\|\theta\|_2^2 = \|\hat{\theta} + v\|_2^2 = \|\hat{\theta}\|_2^2 + 2\langle \hat{\theta}, v \rangle + \|v\|_2^2 = \|\hat{\theta}\|_2^2 + \|v\|_2^2 \geq \|\hat{\theta}\|_2^2,$$

with equality iff $v = 0$. Thus $\hat{\theta}$ uniquely minimizes $\|\theta\|_2$ over all θ satisfying $X\theta = y$.

(e) Example. As in earlier problems, $\hat{\theta} = [0, 1, 1]^T$ and $X\hat{\theta} = y$. Any other feasible θ has the form $\hat{\theta} + v$ with $v \in \text{Null}(X)$, and $\|\hat{\theta} + v\| \geq \|\hat{\theta}\|$.